INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

(School of University of Engineering and Management, Kolkata)

End Semester Examination, November - December, 2025

Degree: **B.C.A.**, Semester: **Third**

Stream: **B.C.A.**

Paper Code: **BCACC301**

Paper Name: **Design and Analysis of Algorithms**

Full Marks: **100** Duration: **3 Hours**

Part - A
Attempt any 10 questions out of 15
questions Each question carries 2 Marks

1. Define space complexity and its importance. **2**

2. What is the time, and space complexity of the following code: **2**

```
a = 0
b = 0
for i in range(N):
   a = a + random()
for i in range(M):
  b= b + random()
```

3. Solve the following recurrence relation: **2**

$T(n) = 2T(n/2)+n \log n$

4. Define job sequencing problem. **2**

5. What is pivot selection in Quick Sort? **2**

6. What is the minimum number of comparisons required in an array to find the average of 10 integers? **2**

7. Define job deadline in job sequencing problem. **2**

8. Write a real-world example of greedy approach. **2**

9. Define DFS and BFS. **2**

10. What is the complexity of Dijkstra's algorithm using adjacency matrix? **2**

11. What is graph coloring? **2**

12. State one difference between connected and bi-connected components. **2**

13. Differentiate between stable and unstable sorting. **2**

14. What is a pseudo-code? **2**

15. Differentiate between NP-hard and NP-complete. **2**

Part - B
Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16. Present an algorithm that searches an unsorted array a[l: n] for the element x. If x occurs, then return a position in the array; else return zero. **5**

17. Solve $T(n) = 7T(n/2) + 18n^2$ using Master's theorem and recursion tree, compare results. **5**

18. You are given a set of *n* jobs. Associated with each job *i* is a processing time t_i and a deadline d_i by which it must be completed. A feasible schedule is a permutation of the jobs such that if the jobs are processed in that order, then each job finishes by its deadline. Define a greedy schedule to be one in which the jobs are processed in non-decreasing order of deadline. **5**

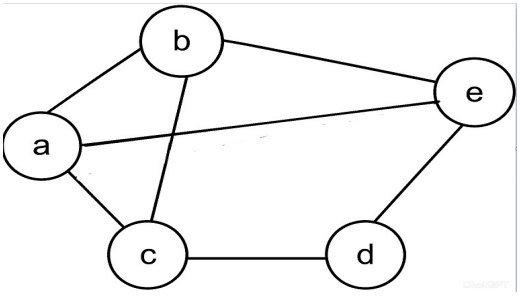
19. Show that for every n such that n mod (k - 1) = 1, there exists a k-ary tree T with n external nodes (in a k-ary tree all nodes have degree at most k). **5**

20. Describe Floyd–Warshall algorithm for all-pairs shortest path problem. **5**

21. Write an algorithm to compute matrix multiplication directly using the standard formula. Show that the number of multiplications needed by your algorithm is mnp where dimensions of A, B, and C respectively be m x n, n x p, and m x p. **5**

22. Solve a Travelling Salesman Problem instance using dynamic programming. **5**

23. Apply Backtracking Method to solve the 3-colouring problem for the following graph. **5**



24. Explain Cook's theorem and its significance in complexity theory. **5**

Part - C
Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25. Circuit Satisfiability Problem is NP-Complete. Discuss this statement. **10**

26. If we want to prove a problem X is NP-Hard, we take a known NP-Hard problem Y and reduce it to X. Discuss this statement. **10**

27. Solve Travelling Salesman Problem for a given cost matrix using dynamic programming and show table filling. **10**

28. a) Explain FIFO Branch and Bound method with example. (5) **10**

b) Explain LC Branch and Bound method with example. (5)

29. a) Derive an algorithm for solving the 0/1 Knapsack Problem using Backtracking Approach. (5) **10**

b) Consider knapsack problem : n = 8. (W1, W2, W3, W4, W5, W6, W2, W8) = (1, 11, 21, 23, 33, 43, 45, 55), P = (11, 21, 31, 33, 43, 53, 55, 65), m = 110. Solve the problem using backtracking approach. (5)

30. Analyze Prim's and Kruskal's algorithms for MST in terms of time complexity and real-world suitability. **10**

31. Write Quick Sort algorithm. Analyze the best case and worst case time complexity of the algorithm. Perform the PARTITION operation once on the following array as per the requirement of the QuickSort Algorithm, assuming the last element of the array to be the pivot element.arr = { 2,8,7,1,3,5,6,4 }. **10**

32. Explain the chained matrix multiplication with suitable example. **10**

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